

Project Title

AI Resume Analyzer Platform

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Course No: CSC 470

Course Title: Software Engineering Lab

Section: D

Semester: Summer 2026

Date of submission: 19/08/2026

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Part A: Evaluating Modern Tool Usage in the Project

1.1 Project Overview

The AI Resume Analyzer Platform is a web application that helps a job seeker improve a resume before applying for a job. The user uploads a resume as a PDF or DOCX file of up to 5 MB and pastes the job description of the post they want. The system reads the resume, sends it with the job description to an AI model, and shows an ATS match score between 0% and 100%. It also gives written feedback in four groups, which are skills, content quality, structure and writing style, and it lists the important keywords that are missing from the resume.

The project is divided into a Backend and a Frontend folder. The backend is an ASP.NET Core 8 Web API written in C#. It uses ASP.NET Core Identity with JWT tokens for login, Entity Framework Core 8 with SQL Server for data, and Swagger for testing the endpoints. The frontend is a React 19 application built with Vite, styled with Tailwind CSS, using Zustand for state management and Axios to call the API.

For the AI part we use Llama 3, an open source model from Meta. It runs on our own computer through Ollama at the address <http://localhost:11434>, and the API talks to it using Microsoft Semantic Kernel. Because the model is local, no resume ever leaves our machine.

1.2 Project Objectives

- To give job seekers free and quick feedback on their resume.
- To show how well a resume matches a specific job description, as a score.
- To point out the skills and keywords that are missing from the resume.
- To keep a history of past analyses so the user can see progress over time.
- To give administrators a safe way to manage users and watch system activity.

1.3 Project Scope

The system covers five modules. Users can register and log in, and the system creates a session that carries the role of User or Admin. Users can upload a resume, see a preview image of the first page, and let the system extract the text. Users can then run the AI analysis and receive the ATS score, the feedback and the missing keywords. The dashboard shows all past analyses with a score over time chart and a side by side comparison. Administrators get a separate dashboard with system statistics, a searchable user list, account status control and system logs.

1.4 Significant Functionalities

The main functions of the system are user registration and login with a JWT token that carries the role, resume upload with file type and size checking, text extraction and preview image creation, AI based scoring against a job description, feedback in four categories, missing and matched keyword lists, analysis history with a progress chart, and an admin dashboard for accounts and statistics. The API is organised into six controllers: Auth, Resume, Ai, Dashboard, JobProfile and Admin.

1.5 Modern Tools and Techniques

We used five modern tools while building the project. The table below shows what each tool was used for and the problem we faced with it.

Tool	Why we used it	Problem we faced
ASP.NET Core 8 Web API with Identity and JWT	Six controllers serve the frontend. Identity creates the account, and a JWT token carries the user role to the API	Middleware order matters: UseAuthentication must come before UseAuthorization. Also, the [Authorize] attribute is per controller, so one controller can be left open by mistake
Entity Framework Core with SQL Server	Code first migrations build the Users, Resumes and Analyses tables and keep them linked	Every model change needs a new migration; we ended with five. Feedback and keywords are stored as JSON text, which is quick to save but cannot be searched with SQL
Llama 3 through Ollama and Semantic Kernel	One prompt returns the ATS score, matched and missing skills and all four feedback groups. The model runs locally, so no resume leaves our machine	Llama 3 sometimes adds text or backticks around the JSON. We had to cut the reply between the first and last brace and add a fallback result. Speed also depends on the load of the machine
iTextSharp and Magick.NET (NuGet packages)	iTextSharp reads the text out of a PDF and Magick.NET renders the first page as a PNG preview at 150 DPI	Text from a two column resume comes out mixed, and a scanned PDF gives no text because it is

		really an image. Preview rendering also costs CPU time
React 19, Vite, Tailwind CSS and Zustand	Builds the user interface. Zustand keeps the login state and the analysis result in one place, and Axios calls the API	The API and the frontend run on different ports, so we had to add a CORS policy on the server before the browser would accept any answer

1.6 Understanding the Limitations

ASP.NET Core makes login easy, but the order of the middleware must be correct. UseAuthentication must come before UseAuthorization, otherwise the server does not know who the user is when it checks the permission. In our Program.cs the three steps are written in a fixed order, CORS first, then authentication, then authorization, and we added a comment there so nobody changes it by mistake.

A second problem with security was our own habit. During development we commented out the [Authorize] attribute on several controllers so that testing with Swagger would be faster. This is easy to forget, so every controller has to be checked again before release. Section 4.2 explains how we handled this.

Entity Framework Core saves a lot of time, but it hides how much data a query really loads, and every change to a model needs a new migration. Our database schema changed five times during the project, so we have five migration files. We also store the category scores, the missing keywords and the detailed feedback as JSON strings inside one table. This was quick to build, but it means we cannot search inside those fields with a normal SQL query.

Llama 3 gives us free and private analysis, because the resume never leaves our machine. But it does not always reply in a clean format, and the speed depends on the load of the machine. The AI reply problem was the hardest one, because it did not happen every time. It only happened with some resumes, so we did not notice it early. Our solution was to check the reply first and show an error message if it is not correct, instead of letting the page break. This check is the decision box named “Analysis successful?” in our activity diagram for Ai Scoring Module.

```
// AiScoringService.cs — keep only the JSON part of the reply
if (responseText.Contains("{}"))
{
    int start = responseText.IndexOf('{');
    int end = responseText.LastIndexOf('}');
    responseText = responseText.Substring(start, end - start + 1); }
```

```
try { return JsonSerializer.Deserialize<AtsAnalysisResultDto>(responseText, options); }
catch { return new AtsAnalysisResultDto { AtsScore = 0,
    Summary = "Failed to evaluate ATS score. Please try again." }; }
```

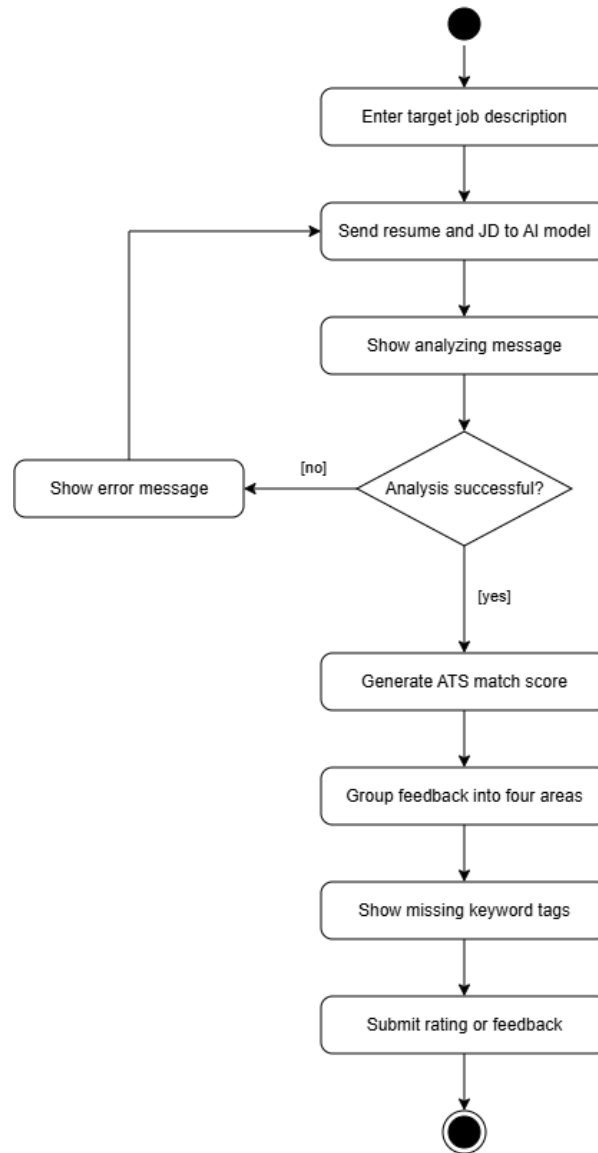


Figure 1: Activity diagram of Ai Scoring Module. The decision box stops a failed analysis from being rated.

Document processing has the clearest limits. We use iTextSharp to pull the text out of a PDF and Magick.NET to render the first page as a PNG preview at 150 DPI. A resume written in two columns sometimes gives mixed up text, and a scanned PDF gives no text at all, because the file is really an image. We planned to accept .txt files as well, but the upload endpoint currently allows only .pdf and .docx. This difference between the plan and the code was found during this review and has been noted for the next iteration.

Part B: Assessing Sustainability in the Engineering Solution

2.1 Environmental Practices

1. One AI request instead of many

One prompt asks Llama 3 for the score, the matched skills, the missing skills, the strengths, the improvements and all four feedback groups together, and everything is read from that single reply. If we asked the model separately for each part, the same analysis would need about four times more computing power and electricity.

2. Fewer database queries

At first the dashboard ran one query for every saved analysis to draw the score chart. We changed it to one query that returns only the score and the date. For a user with twenty analyses, twenty database calls became one, which lowers the work done by the server.

3. Checking the file before doing any work

The upload endpoint checks the size against a 5 MB limit and the extension against an allowed list before anything else runs. A wrong file is rejected at once, so no parsing, no preview rendering and no AI work is wasted on a file we will not use.

4. Doing heavy work only once

The extracted text is saved in the database and the preview image is written to disk one time. So the same PDF is never parsed twice and the same preview is never rendered twice, even if the user opens the analysis again later.

5. No automatic analysis

The analysis does not start by itself after upload. The user must click Analyze Resume, so the system never spends energy on work that nobody asked for.

2.2 Impact Analysis

The positive impact is that a student or a job seeker can improve a resume for free. They do not need to print many drafts or pay for a resume review service, and they can compare an old analysis with a new one instead of starting again. Digital feedback also reaches people who cannot visit a career office.

The negative impact is that AI work uses much more electricity than a normal web page. Even our own Llama 3 model needs a lot of processing for one analysis, and rendering a preview image at 150 DPI also costs CPU time. Saving every uploaded resume and every result also uses more disk space as time passes, and the servers and computers behind the system create electronic waste at the end of their life.

Our main trade-off was between quality and energy. A larger version of Llama 3 gave better feedback, but it was slow and used more power, and it did not meet our 15 second target for one analysis. We chose a smaller version and improved the instructions we send to it instead. This kept the result useful while keeping the cost of one analysis low.

Part C: Analyzing Societal Impact and Responsibilities

3.1 Contextual Analysis

A. Legal side: personal data protection

A resume contains a name, phone number, email address and job history, which is personal information. Passwords are never stored directly: ASP.NET Core Identity hashes them for us, and our password rules require a digit, a lowercase letter and at least six characters. Uploaded files are also saved under a random GUID name instead of the original file name, so one user cannot guess the address of another user's resume. An administrator can see the user list and the system statistics without needing to open another person's resume.

B. Social side: the score affects a real person

An ATS score can decide whether someone keeps or throws away a resume, so a wrong or unclear score can affect a person's career. For this reason we never show only a number. The score is always shown together with the four feedback groups and the missing keywords, so the user can see the reason behind it and treat it as advice, not as a final decision.

C. Cultural side: resume styles are different in different countries

In some countries a photo or a date of birth is normal in a resume, and in other countries it is not accepted. So our feedback is about skills, structure and writing quality only. The system does not give a low score just because a resume follows a local style.

D. Safety and reliability

The system is not safety critical, but a wrong result still misleads the user. So the AI reply is checked before it is shown, error messages are clear, and every analysis row keeps a foreign key to both its user and its resume, so the records stay correctly linked.

E. Privacy by design: the model stays local

Because Llama 3 runs on our own machine through Ollama, the resume text is never sent to an outside company. If we had used a paid cloud AI service, every resume would leave our control and we would have to explain that to the user. Running the model locally removed that problem completely.

3.2 Responsibilities

First, we are responsible for the data users trust us with. A resume is uploaded for one purpose only, so every analysis stays connected to its own user account and is never shown to another account.

Second, we are responsible for being honest about the system. The score comes from an AI model and it can be wrong. So we describe it as help for improving a resume, not as a promise that the person will get the job.

Third, we are responsible for keeping the admin features safe. Banning an account or reading system statistics is a powerful action, so the role must be checked by the server on every request. Hiding the admin link in the React interface is not enough, because anyone can call the API directly with a tool such as Swagger or Postman. In our swim lane diagram for Module 5, the role check is the first step in the System lane for exactly this reason, and Section 4.2 describes how we applied it in the code.

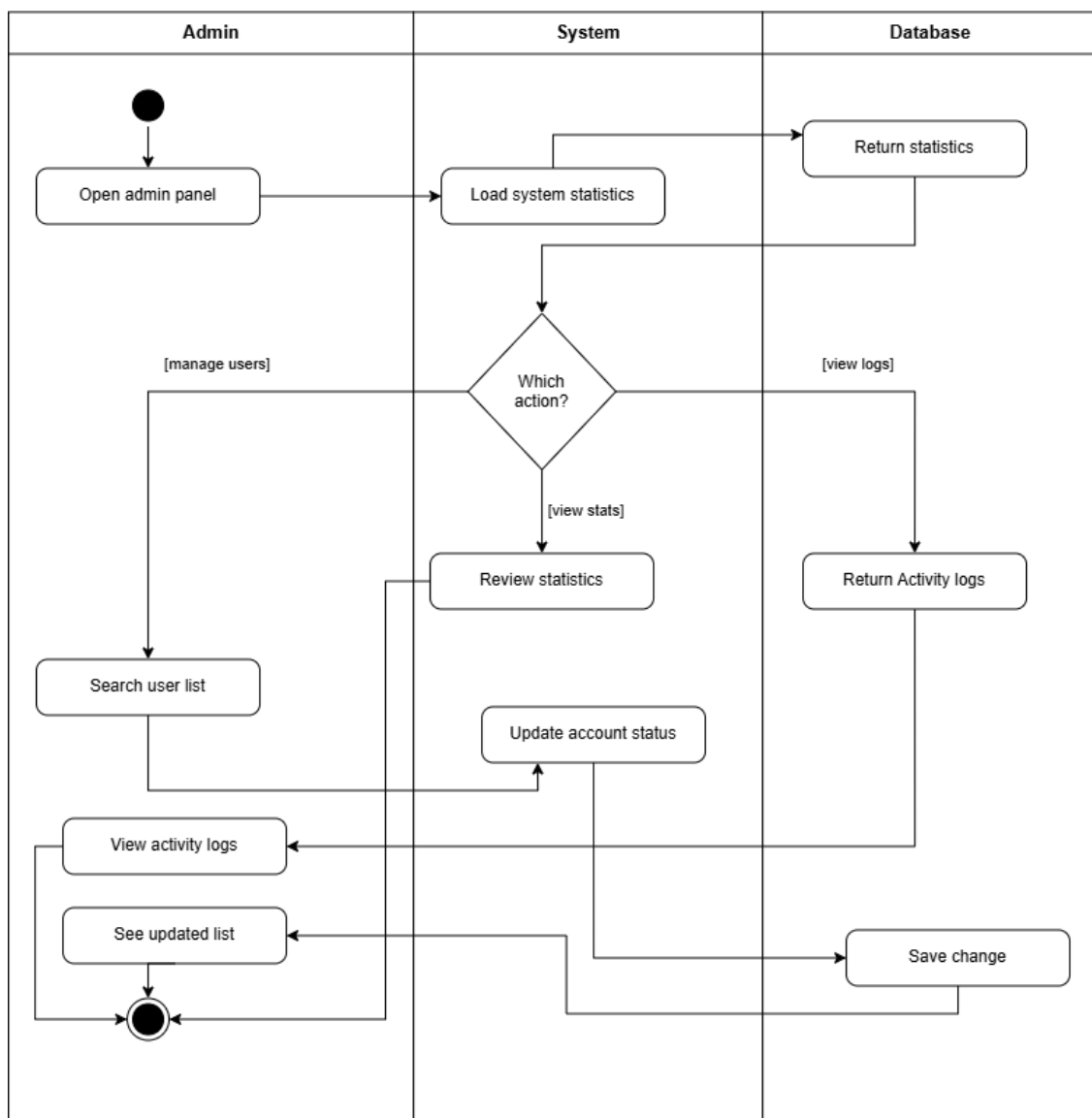


Figure 2: Swim lane diagram of Admin panel Module. The role check comes before any admin work.

Part D: Exploring Ethical Principles in Professional Engineering Practice

4.1 Ethical Considerations

1. Working with real resumes

To test the system we needed real resumes. We used our own resumes and sample resumes with the personal details removed. We did not collect resumes from classmates who had not agreed to give them.

2. Being clear about an automated decision

It was easy to show the ATS score as a final number. We decided to show the reason with it instead, so the user understands why the score is high or low. This was an ethical choice, not a technical one.

3. Being honest about what the system cannot do

The version of Llama 3 we run is small, so the quality of the feedback is not always the same, and a scanned PDF gives no text at all. We did not test or present the project only with the resumes that we knew would give a good result.

4.2 Ethical Decision-Making

The clearest problem was the deadline against security. While building the API we commented out the [Authorize] attribute on the Admin, Dashboard, Ai and JobProfile controllers, so that we could test the endpoints in Swagger without a token. This saved time every day, but it meant those endpoints were open to anyone who knew the address. We left a note in the code to turn them back on later, and during the review before submission we came back to this decision.

We compared the two sides. Turning the checks on costs testing time and could break the demonstration if a token is missing. Leaving them off looks fine in the demonstration, but it means an unknown person could read the user list or the statistics of real accounts. The risk of the shortcut falls on the users and not on us, so we treated the security check as part of finishing the feature, not as an extra. The rule we agreed on was that a feature is not complete until its access control is enabled and tested.

We used the same thinking for the AI reply problem. When Llama 3 returns text that is not valid JSON, our service could quietly return an average score and the page would still look complete. We chose instead to return a score of zero with the message “Failed to evaluate ATS score. Please try again.”, because a made up score would give the user wrong information about their own resume. In both cases our rule was simple: if a shortcut helps us but the risk goes to the user, it is not a good shortcut.

Conclusion

The AI Resume Analyzer Platform shows how software engineering ideas can be applied to a real problem faced by students and job seekers. The project brings together an ASP.NET Core 8 Web API, Entity Framework Core with SQL Server, a React and Zustand frontend, and a self hosted Llama 3 model reached through Semantic Kernel and Ollama, along with requirement analysis, UML analysis modelling, layered design and role based security. While building it we also looked beyond the code: we reduced the computing work needed for one analysis, we protected personal data, we made the score explainable instead of hiding it behind a number, and we chose security over a faster demonstration. For this reason the project is not only a working software solution but also an example of responsible engineering practice.

References

- Microsoft. ASP.NET Core Documentation. Microsoft Learn. learn.microsoft.com/aspnet/core
- Microsoft. Entity Framework Core Documentation. Microsoft Learn. learn.microsoft.com/ef/core
- Ollama. Ollama Documentation. ollama.com/docs
- Microsoft. Semantic Kernel Documentation. learn.microsoft.com/semantic-kernel
- Project repository: github.com/EhetisumSharif/ai-resume-analyzer

Department of Computer Science and Engineering
Bachelor of Computer Science and Engineering (BCSE)

Course (CSC 470) along with
Complex Engineering Problem (CEP) and Complex Engineering Activities (CEA)

- Course Code and Title: CSC 470, Software Engineering Lab
- Semester: Summer 2026
- Student Name: Ehetisum Sharif, Nishat Tasnim, Md. Sajid
- Student ID: 23103380, 23103388, 23103399
- Course Instructor Name: Avijit Biswas, Lecturer, CSE Department, IUBAT
- Which P's are addressed (Requirements: P1 is must and Some or all P2 to P7)?

Name of CEA	Addressed (Explain how it was addressed?) or —	
P1: Depth of Knowledge (Required one or more K from K3, K4, K5, K6, or K8)	K3	Applied software engineering and programming knowledge to build a working system: an ASP.NET Core 8 Web API with six controllers and a React 19 frontend, covering JWT login, file upload, AI analysis, a user dashboard and an admin panel.
	K4	Analysed the manual way of checking a resume, identified the problems, and wrote the requirements as five functional modules with clear performance, security and reliability targets.
	K5	Combined several areas of knowledge: requirement analysis, UML analysis modelling, relational database design with five EF Core migrations, REST API development, document processing with iTextSharp and Magick.NET, local AI integration through Semantic Kernel and Ollama, and token based security.
	K6	Checked feasibility and performance limits (5 second preview, 15 second analysis), chose a local Llama 3 model instead of a paid cloud service, and reviewed the access control on every controller before submission.
	K8	Used official documentation and standard references, listed in



Name of CEA	Addressed (Explain how it was addressed?) or —
	the References section, during development.
P2: Conflicting Requirements	Two conflicts were faced. First, analysis quality against speed and energy: a larger Llama 3 model gave better feedback but broke our 15 second target, so a smaller model with a stronger prompt was chosen. Second, testing speed against security: the [Authorize] checks were commented out during development for faster Swagger testing, and had to be enabled before release.
P3: Depth of Analysis	Each of the modules was analysed with an activity diagram and a swim lane diagram, so both the order of the steps and the responsible actor were identified before coding. Every step is traceable to a stated requirement of the system.
P4: Familiarity of Issues	The project addresses well known problems in job applications: resumes rejected by ATS filters, missing keywords, weak structure, and the cost of a professional resume review.
P5: Extent of Applicable Codes	Personal data handling follows normal data protection practice: password hashing through ASP.NET Core Identity, token based access control, random GUID file names, and a locally hosted AI model so that resume data never leaves the machine.
P6: Extent of Stakeholders	Three stakeholder groups were considered: job seekers who upload resumes, administrators who manage accounts and watch system statistics, and employers whose job descriptions set the standard for the score.
P7: Interdependence	The modules depend on each other: the Resume controller gives the extracted text to the AI service, the AI service fills the Analysis record, the Dashboard controller reads those records, and the Admin controller reads the totals produced by all of them. The Analysis table holds foreign keys to both the user and the resume.

Comments: The AI Resume Analyzer Platform covers the full software engineering process, from requirement analysis and UML analysis modelling to implementation, security and testing. All seven activities from P1 to P7 are addressed. P2 is supported by two documented conflicts, quality against energy and deadline against security, and P5 is supported by data protection practice

Criteria	Excellent (91-100)	Very Good (81-90)	Good (71-80)	Satisfactory (61-70)	Needs Improvement (51-60)	Unsatisfactory (0-50)
Modern Tool Usage	Comprehensive use of modern tools and techniques. Applies a wide range of appropriate engineering and IT tools for complex engineering problems, with an understanding of limitations.	Utilizes a diverse array of modern tools, but with some gaps in application or understanding.	Incorporates several modern tools, but may lack depth or understanding of limitations.	Limited use of modern tools, with a focus on basic techniques and resources.	Relies primarily on traditional tools, with minimal application of modern techniques.	No use of modern engineering or IT tools; relies solely on outdated methods.
K6 - Knowledge of Engineering Practice	Thorough understanding of engineering practice in the discipline. Displays strong technical knowledge in the selected practice areas.	Good knowledge of engineering practice with minor gaps in specific areas.	Basic understanding of engineering practice, with several gaps in technical knowledge.	Limited technical knowledge in engineering practice.	Very limited knowledge in engineering practice.	No understanding of engineering practice in the discipline.
Environment and Sustainability	Thorough evaluation of environmental and sustainability factors. Demonstrates a strong commitment to sustainable engineering practices and considers societal impacts.	Good evaluation of sustainability, but with some missed opportunities for improvement.	Basic understanding of environmental considerations with several oversights.	Minimal focus on sustainability or environmental impacts.	Limited attention to environmental and societal considerations.	No evaluation of sustainability; disregards environmental and societal impacts.
K7 - Role of Engineering in Society	Comprehensive understanding of engineering's role in society. Addresses ethics, public safety, and the broader impacts of engineering activity.	Good comprehension of engineering's societal role, with minor gaps in addressing key issues.	Basic understanding of engineering's societal role, focusing on traditional aspects.	Limited exploration of engineering's societal and cultural implications.	Very limited comprehension of engineering's societal role and related issues.	No comprehension of engineering's role in society or ethical responsibility.
Ethics	Thorough application of ethical principles. Demonstrates a strong commitment to professional ethics and addresses potential ethical dilemmas effectively.	Good understanding of ethical principles, with minor gaps in applying professional ethics.	Basic application of ethical principles, with several areas lacking clarity.	Limited focus on professional ethics or addressing ethical issues.	Very limited consideration of ethics and professional responsibilities.	No application of ethical principles or professional ethics.